

Computer Networks I

IPv4 Datagram Format and Addressing

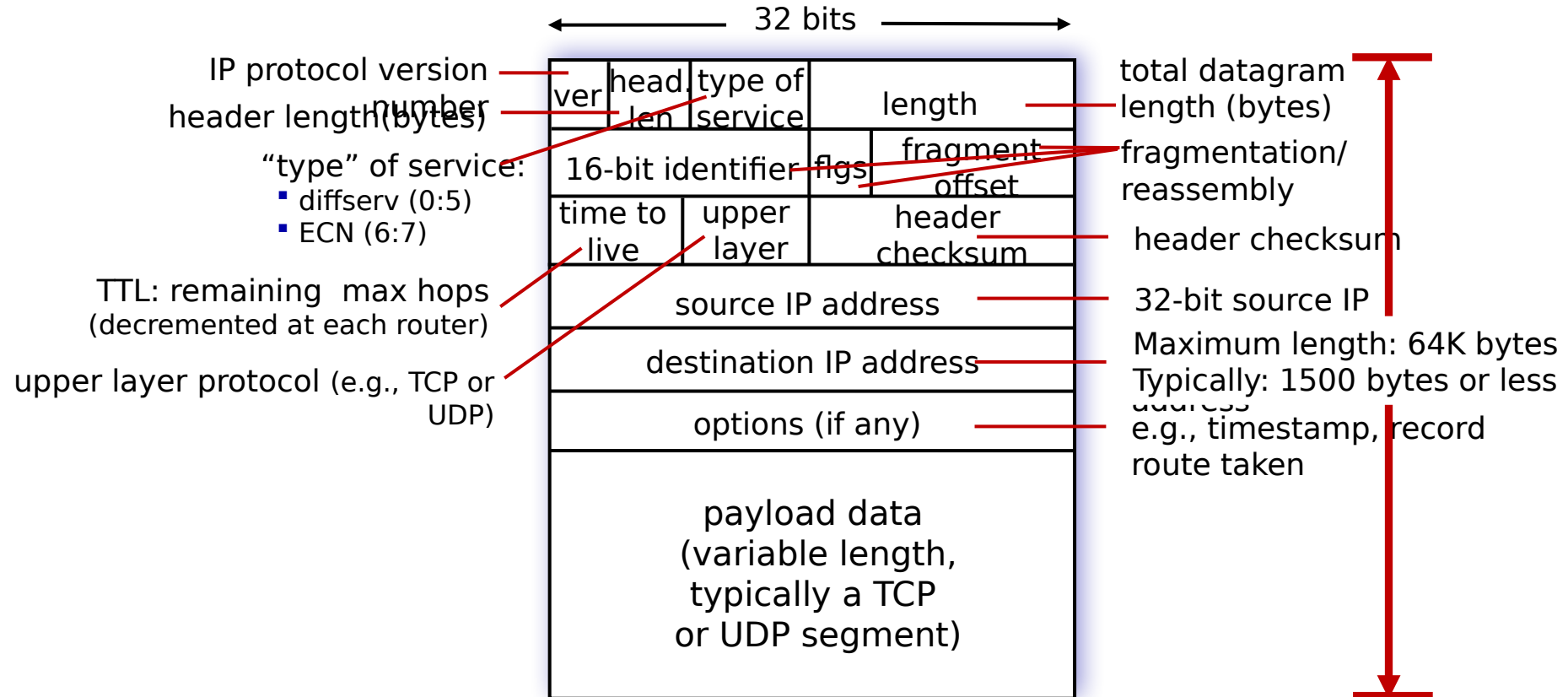
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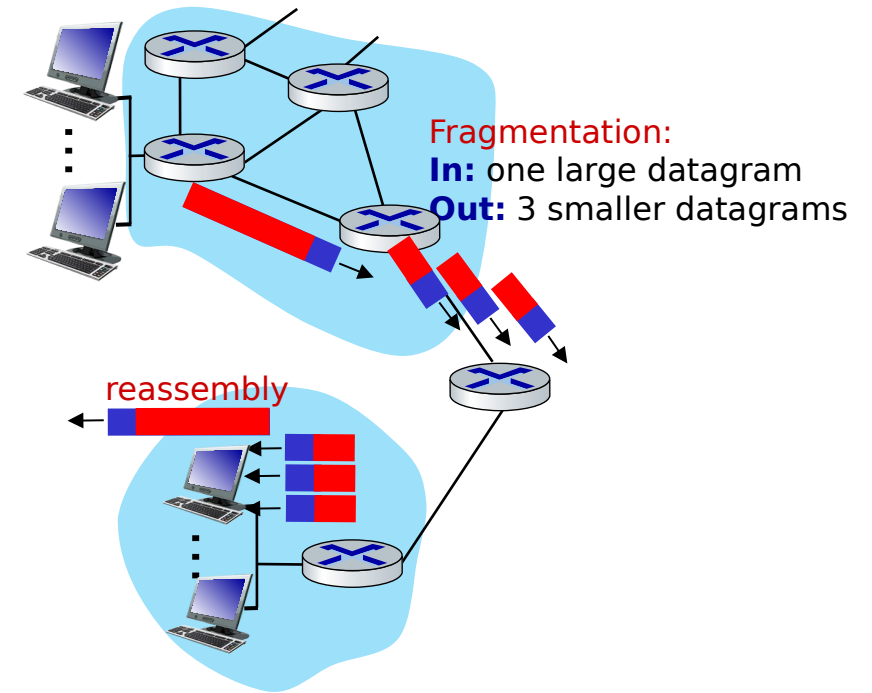
IPv4 Datagram Format

IP Datagram format



IP Fragmentation/Reassembly

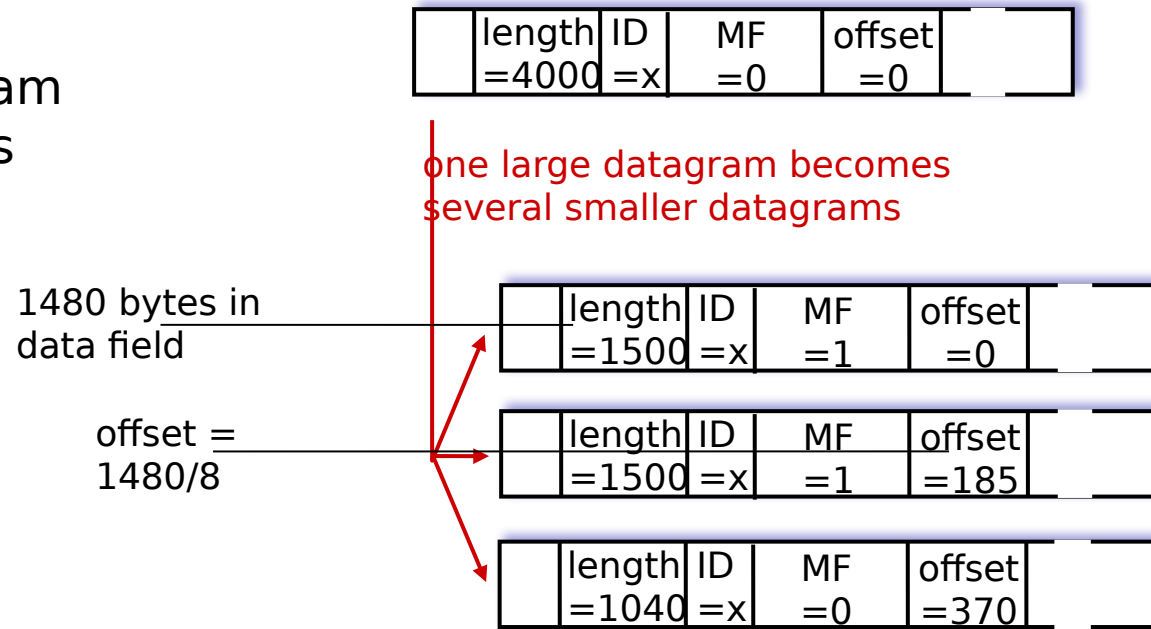
- Network links have MTU (max. transfer size) - largest possible link-level frame
 - Different link types, different MTUs
 - Ethernet 1.5 KB, WiFi 2.3 KB
- Large IP datagram divided (“fragmented”) within net
 - One datagram becomes several datagrams
 - **Reassembled** only at destination
 - IP header bits used to identify, order related fragments



IP Fragmentation/Reassembly

Example:

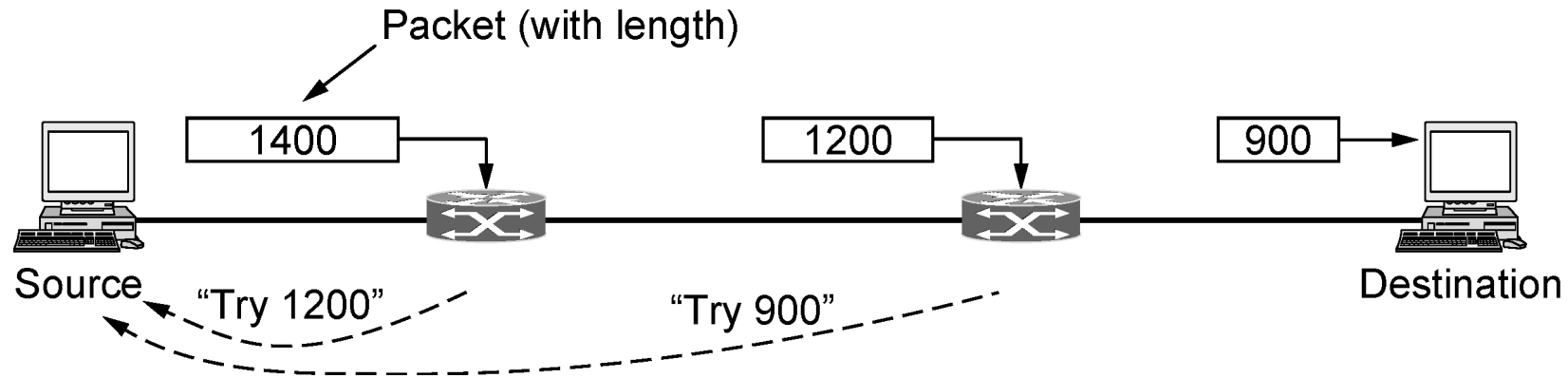
- 4000 byte datagram
- MTU = 1500 bytes



IP Fragmentation/Reassembly

However fragmentation is undesirable

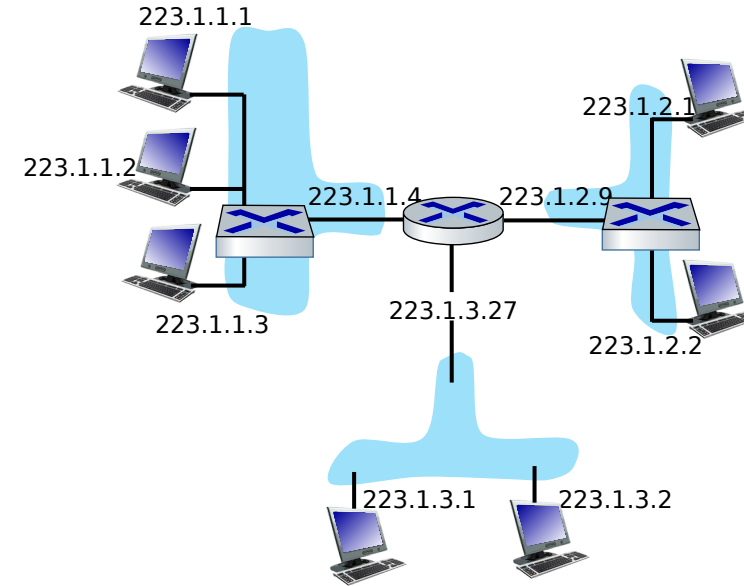
- More overhead for the routers
- **Solution:** discover the MTU that will fit for all the links
 - Implemented using ICMP → DF bit is set to 1



IP Addressing

IP addressing: introduction

- **IP address:** 32-bit identifier associated with each host or router **interface**
- **Interface:** connection between host/router and physical link
 - Router's typically have multiple interfaces
 - Host typically has one or two interfaces (e.g., wired Ethernet, wireless 802.11)

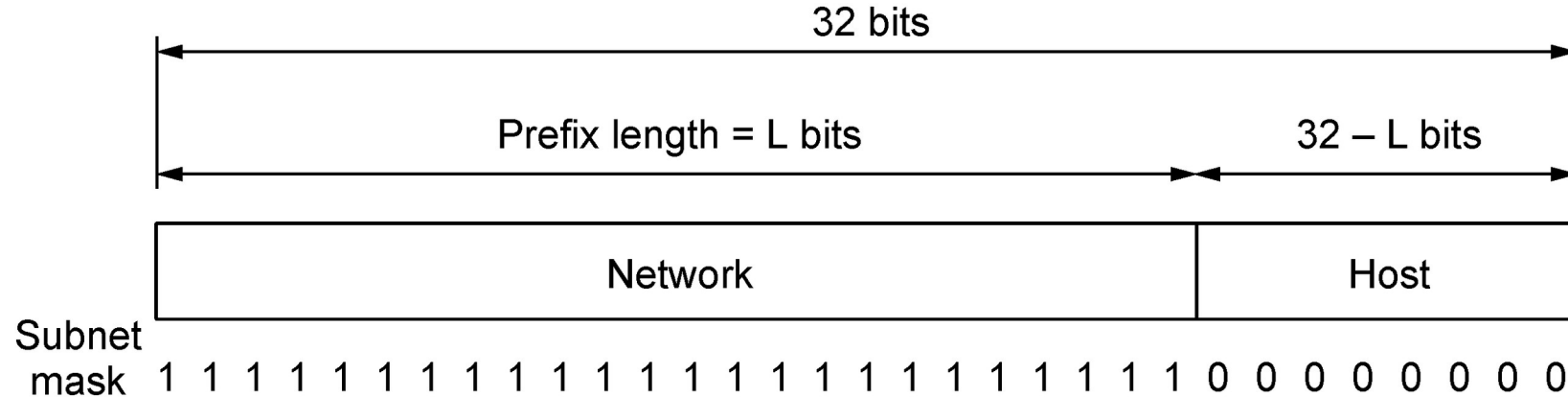


Dotted-decimal IP address notation:

223.1.1.1 = $\underbrace{11011111}_{223} \underbrace{00000001}_{1} \underbrace{00000001}_{1} \underbrace{00000001}_{1}$

Subnets

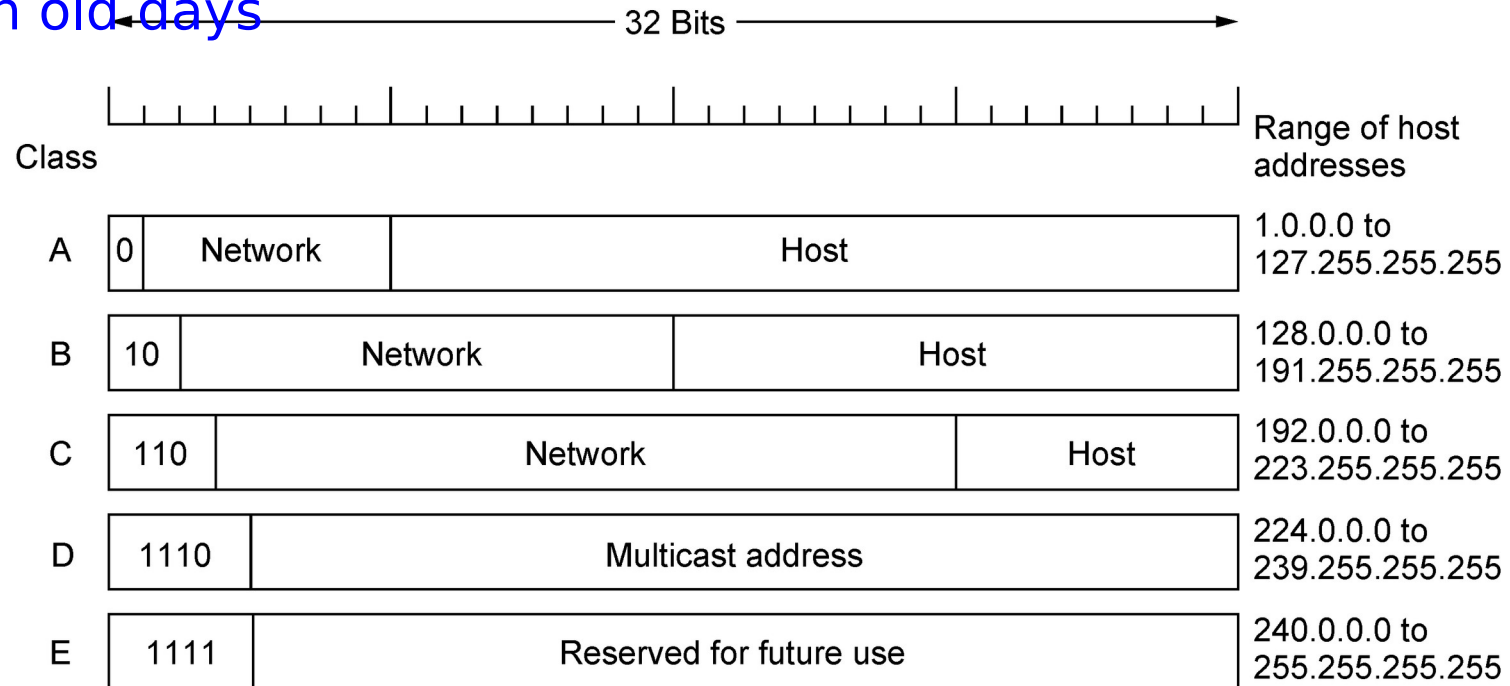
- IP addresses have structure:
 - **Network part:** devices in same network have common high order bits
 - **Host part:** remaining low order bits



IP addressing: Classful Addressing

Classful Addressing: Fixed number of bits for network and host address

- Used in old days



IP addressing: Classful Addressing

Classful Addressing: Fixed number of bits for network and host address

- Used in old days
 - **Network address:** All 0's in the host address part
 - Class-A: 01111111.00000000.00000000.00000000 or 127.0.0.0
 - Class-B: 10101000.10101100.00000000.00000000 or 168.172.0.0
 - Class-C: 11011000.01111011.10010001.00000000 or 216.123.145.0
 - **Broadcast address:** All 1's in the host address part
 - Class-A: 01111111.11111111.11111111.11111111 or 127.255.255.255
 - Class-B: 10101000.10101100.11111111.11111111 or 168.172.255.255
 - Class-C: 11011000.01111011.10010001.11111111 or 216.123.145.255
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IP addressing: Classful Addressing

How many hosts can a Class A, Class B and Class C address accommodate?

- Class-A:

- Class-B:

- Class-C:

A	0	Network	Host	1.0.0.0 to 127.255.255.255
B	10	Network	Host	128.0.0.0 to 191.255.255.255
C	110	Network	Host	192.0.0.0 to 223.255.255.255

Limitation of Classful Addressing

Suppose you have 300 hosts. Which IP address you will go for?

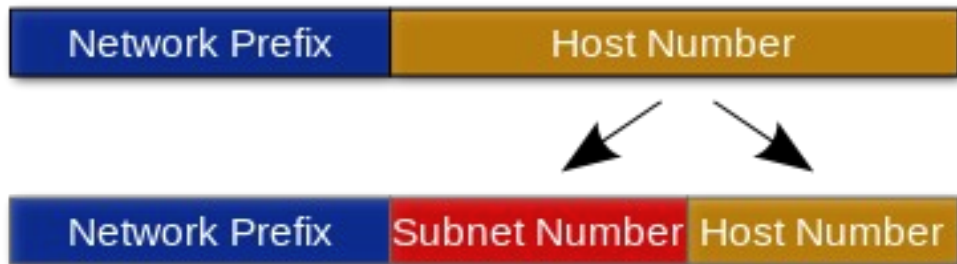
- Class-C: Less addresses
- Class-B: Huge wastage of IP addresses

A	0	Network	Host	1.0.0.0 to 127.255.255.255
B	10	Network	Host	128.0.0.0 to 191.255.255.255
C	110	Network	Host	192.0.0.0 to 223.255.255.255

IP addressing: CIDR

CIDR: **C**lassless **I**nter**D**omain **R**outing (pronounced “cider”)

- Network or subnet portion of address of arbitrary length
- Address format: **a.b.c.d/x**, where x is # bits in network or subnet portion of address

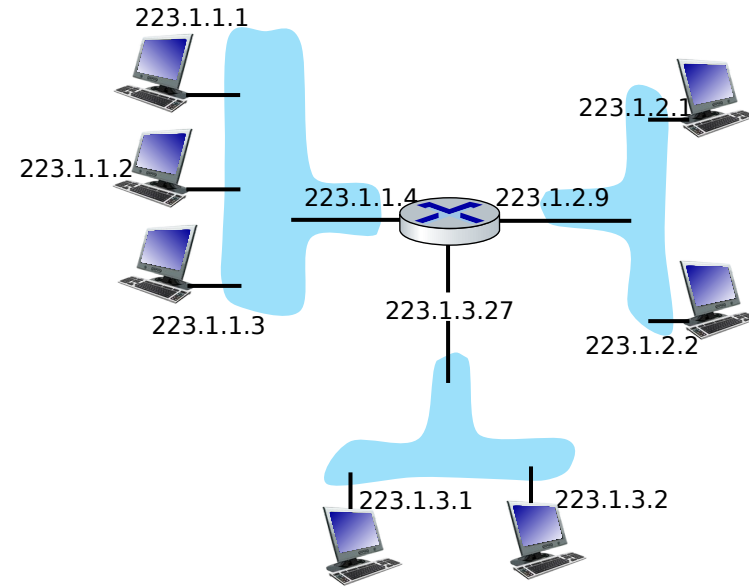


← subnet part → ← host part →
11001000 00010111 00010000 00000000
200.23.16.0/23

Src:
https://commons.wikimedia.org/wiki/File:Subnetting_operation.svg

Subnets

- What's a subnet ?
 - Device interfaces that can physically reach each other **without passing through an intervening router**

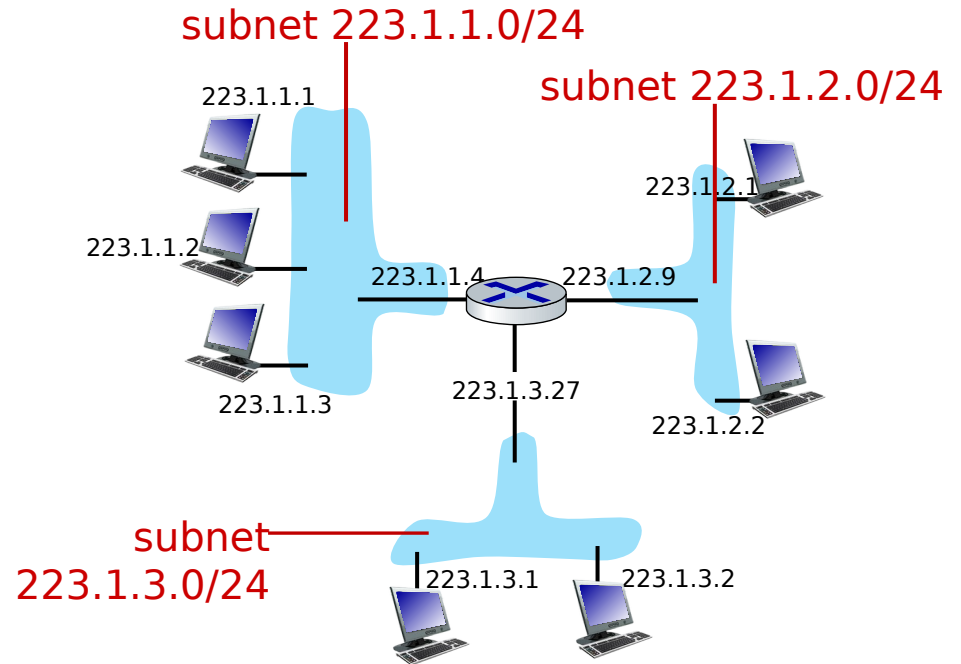


Network consisting of 3 subnets

Subnets

Recipe for defining subnets:

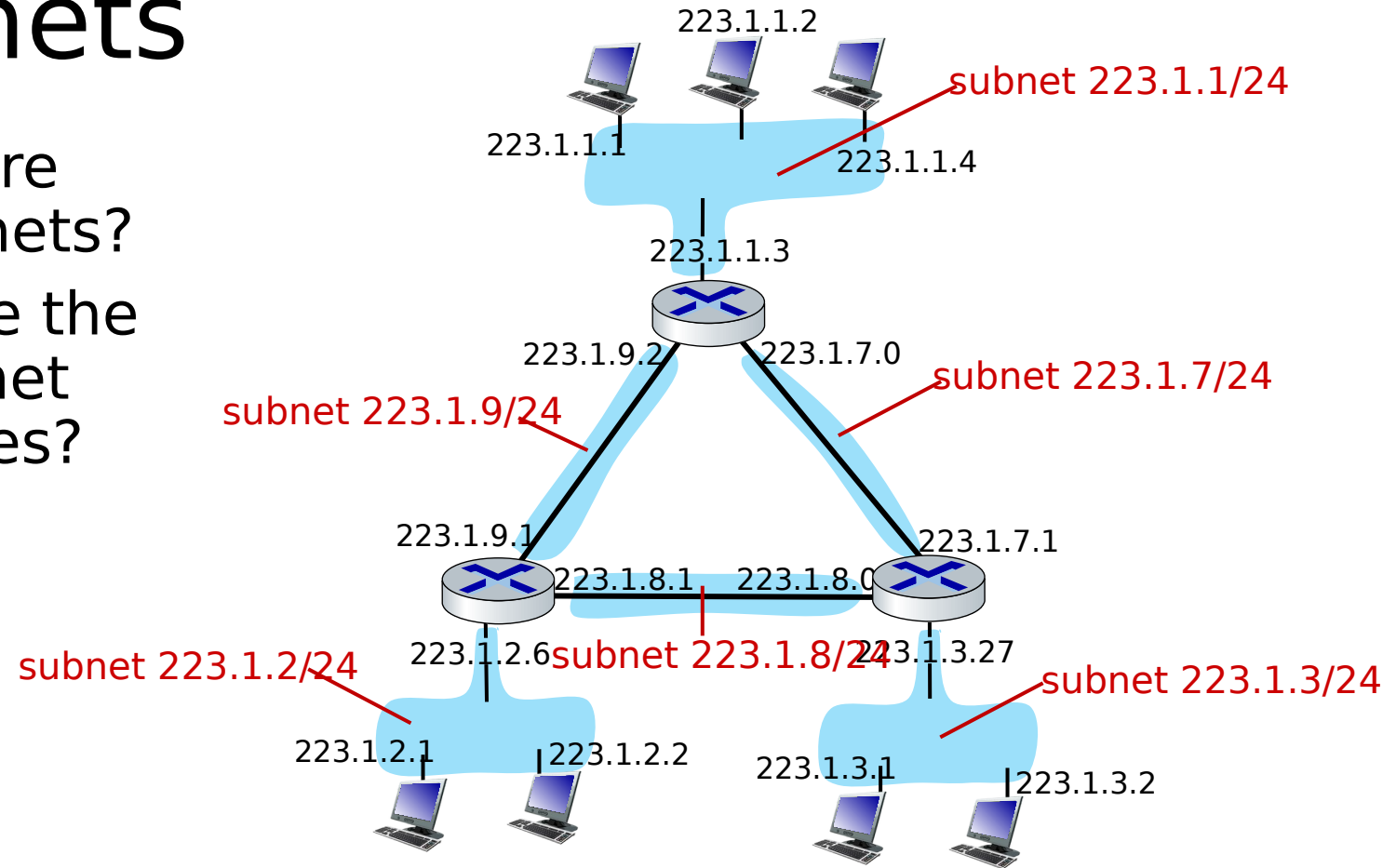
- Detach each interface from its host or router, creating “islands” of isolated networks
- Each isolated network is called a **subnet**



subnet mask: /24
(high-order 24 bits: subnet part of IP address)

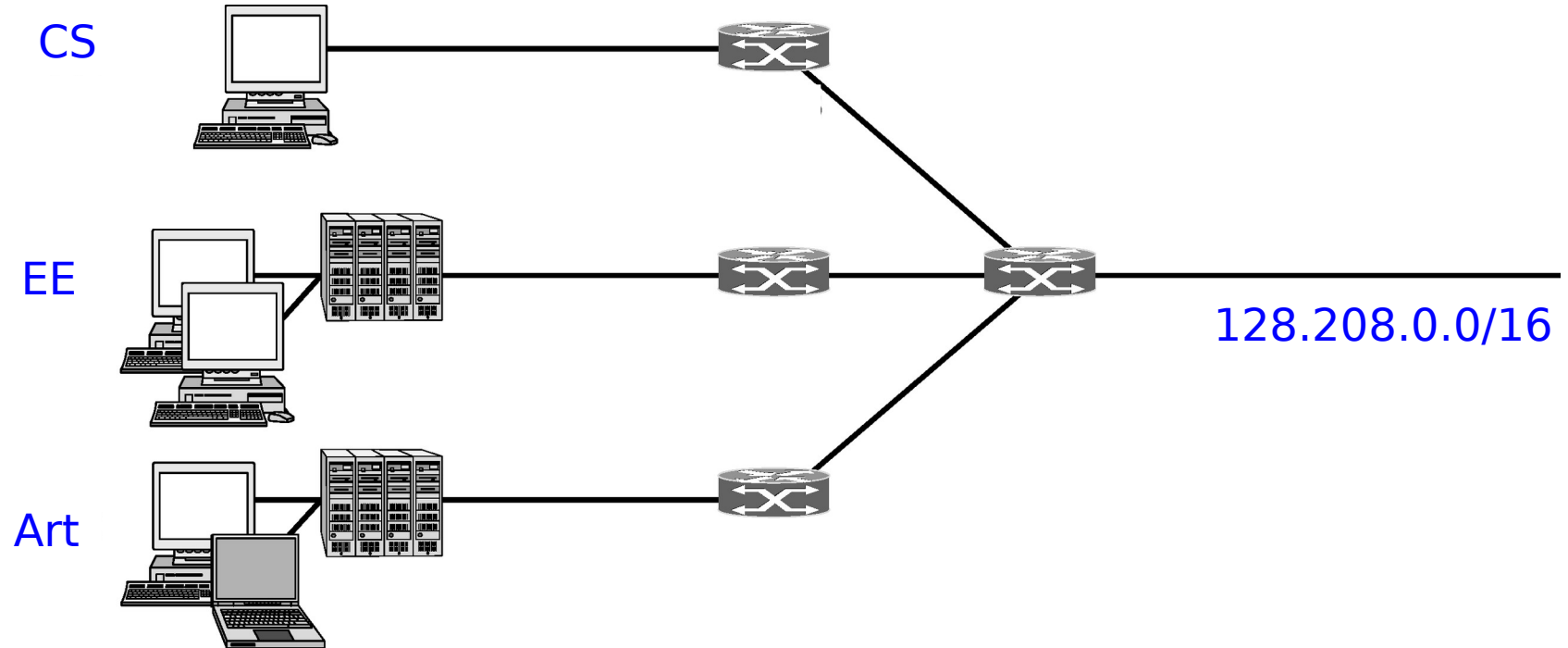
Subnets

- Where are the subnets?
- What are the /24 subnet addresses?



Subnets

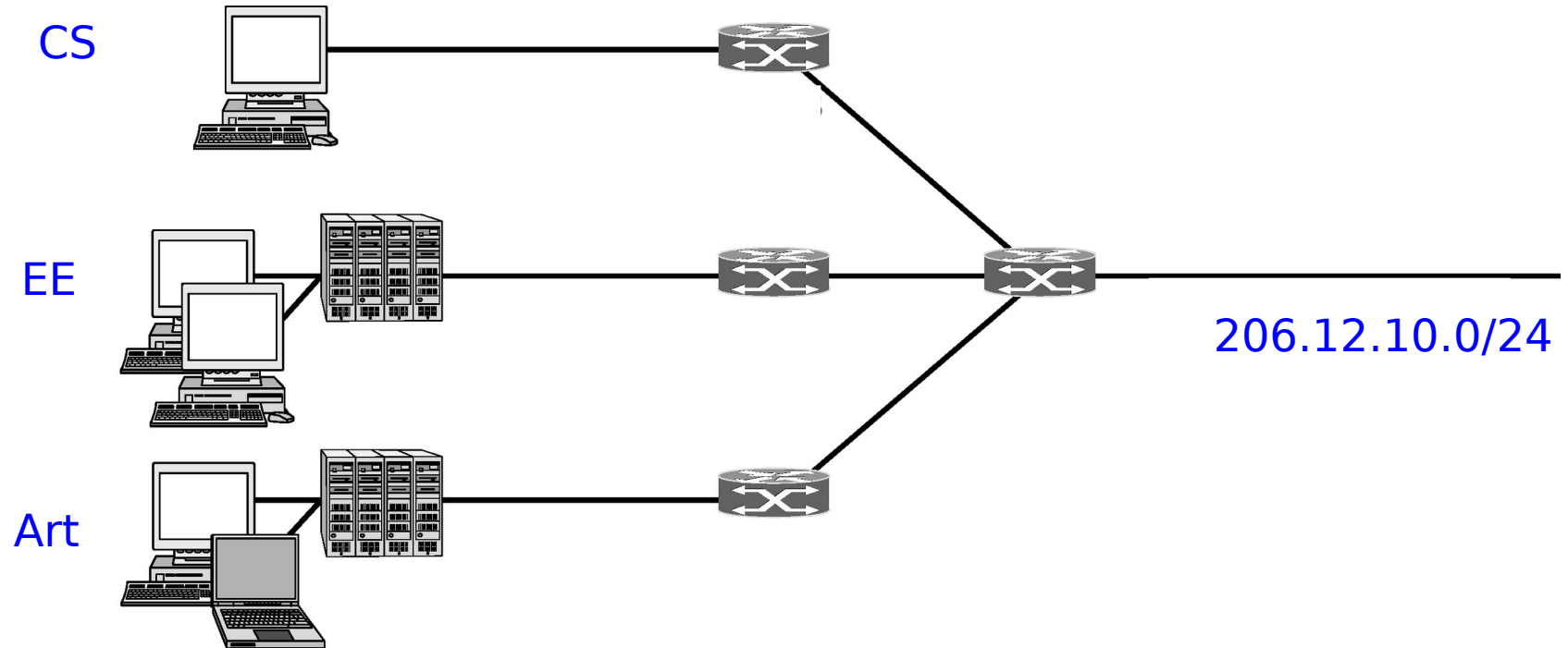
- CS: 550
- EE: 550
- Art: 400



Useful Link: <https://www.cisco.com/c/en/us/support/docs/ip/dynamic-address-allocation-resolution/13711-40.html>

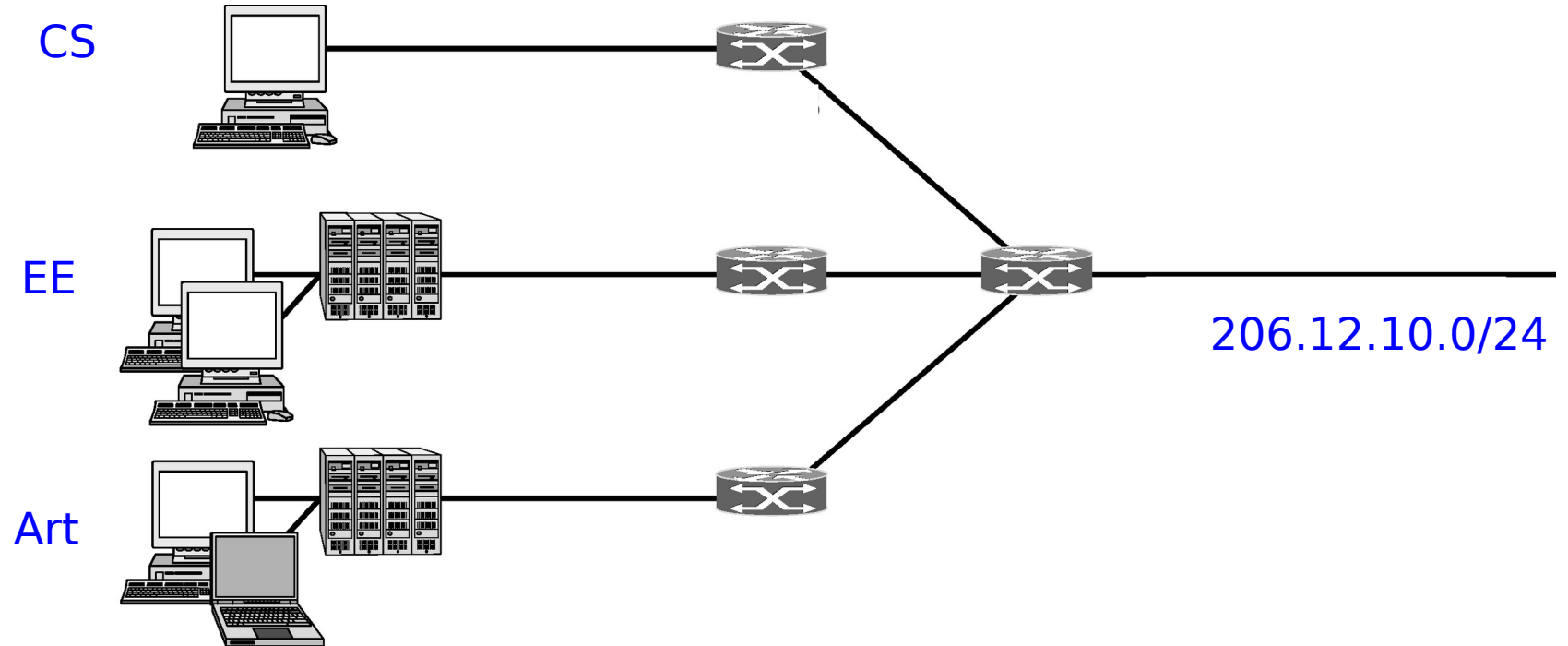
Subnets: Fixed Length Subnetting

- CS: 60
- EE: 10
- Art: 10



Subnets: Variable Length Subnetting

- CS: 60
- EE: 10
- Art: 10



Aggregation of IP Prefixes

University	First address	Last address	How many	Prefix
Cambridge	194.24.0.0	194.24.7.255	2048	194.24.0.0/21
Edinburgh	194.24.8.0	194.24.11.255	1024	194.24.8.0/22
(Available)	194.24.12.0	194.24.15.255	1024	194.24.12.0/22
Oxford	194.24.16.0	194.24.31.255	4096	194.24.16.0/20

New York

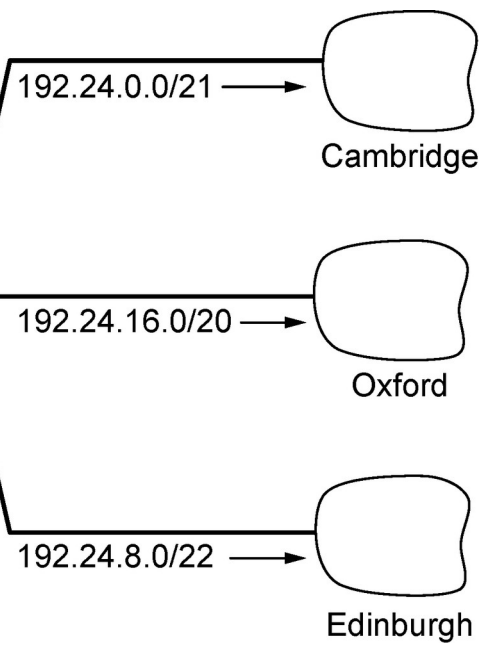


194.24.0.0/19
(1 aggregate prefix)

London



(3 prefixes)



Summary

□ Internet Protocols:

- IPv4 datagram format
 - Fragmentation and Reassembly
 - IP addressing
 - Classful addressing
 - Classless InterDomain Routing
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